
Algorithm of HOIF estimators for ATE

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Contents

1 Input data	1
2 Target formula	1
3 The Integrated Algorithm (Main Interface)	2
4 Step-by-Step Technical Notes	3
4.1 Transformation of Covariates	3
4.2 Compute the Residuals	3
4.3 Compute the Inverse of the Weighted Gram Matrix	3
4.4 Compute the Projection Matrix	4
4.5 Compute the HOIF Estimators	4
Bibliography	5

1 Input data

Let the observed data be denoted by $(X_i, A_i, Y_i)_{i=1}^n$, where $A_i \in \{0, 1\}$ is the binary treatment indicator, $Y_i \in \mathbb{R}$ is the observed outcome, and $X_i \in \mathbb{R}^p$ represents the vector of covariates.

We assume the availability of pre-computed nuisance function estimators: the conditional mean outcomes $(\hat{\mu}(1, X_i), \hat{\mu}(0, X_i))$ and the propensity score $\hat{\pi}(X_i)$. All these estimators map to the real line $\mathbb{R}$.

2 Target formula

The core function `hoif_ate()` in this R package is to compute the below estimators.

Let $\hat{\psi}_a^{\text{AIPW}}$ denote the standard first-order doubly robust / double machine learning / AIPW estimator of $\psi_a = \mathbb{E}[Y(a)]$, and $\widehat{\text{ATE}}^{\text{AIPW}} = \hat{\psi}_1^{\text{AIPW}} - \hat{\psi}_0^{\text{AIPW}}$. Conditional on the estimated nuisance functions, this estimator is biased.

BC_m^{ATE} is the m -th order higher order influence function (HOIF) estimator of the **estimable bias** of the AIPW estimator of the average treatment effect (ATE) in causal inference, developed in a series of works by James M. Robins and his collaborators [1], [2], [3], [4]. It is **not** itself an estimator of the ATE; the sign convention is such that adding it to the AIPW estimate removes the estimable bias:

$$\widehat{\text{ATE}}_m = \widehat{\text{ATE}}^{\text{AIPW}} + \text{BC}_m^{\text{ATE}}$$

(In the output of `hoif_ate()`, the components `H0IF1`, `H0IF0` and `ATE` correspond to HOIF_m^1 , HOIF_m^0 and BC_m^{ATE} ; the component name `ATE` is kept for backward compatibility – it holds the **ATE bias correction**, not the ATE itself.)

$$\begin{aligned}
\text{BCATE}_m &= \text{HOIF}_m^1(\hat{\Omega}^1) - \text{HOIF}_m^0(\hat{\Omega}^0) \\
\text{HOIF}_m^a(\hat{\Omega}^a) &= \sum_{j=2}^m \mathbb{IF}_j^a(\hat{\Omega}^a) \\
\mathbb{IF}_m^a(\hat{\Omega}^a) &= (-1)^m \frac{(n-m)!}{n!} \sum_{(i_1, \dots, i_m) \in I_1^{\otimes m} : i_1 \neq i_2 \neq \dots \neq i_m} r_{i_1}^a Z_{i_1}^\top \hat{\Omega}^a \prod_{s=2}^{m-1} \left\{ (Q_{i_s}^a - (\hat{\Omega}^a)^{-1}) \hat{\Omega}^a \right\} s_{i_m}^a Z_{i_m} R_{i_m}^a \\
&= \sum_{j=2}^m \binom{m-2}{m-j} \mathbb{U}_j^a(\hat{\Omega}^a) \\
\mathbb{U}_m^a(\hat{\Omega}^a) &= (-1)^m \frac{(n-m)!}{n!} \sum_{(i_1, \dots, i_m) \in I_1^{\otimes m} : i_1 \neq i_2 \neq \dots \neq i_m} r_{i_1}^a Z_{i_1}^\top \hat{\Omega}^a \prod_{s=2}^{m-1} \{ Q_{i_s}^a \hat{\Omega}^a \} s_{i_m}^a Z_{i_m} R_{i_m}^a \\
&= (-1)^m \frac{(n-m)!}{n!} \sum_{(i_1, \dots, i_m) \in I_1^{\otimes m} : i_1 \neq i_2 \neq \dots \neq i_m} r_{i_1}^a \prod_{s=1}^{m-1} \{ Z_{i_s}^\top \hat{\Omega}^a s_{i_{s+1}}^a Z_{i_{s+1}} \} R_{i_m}^a
\end{aligned}$$

Where

$$\begin{aligned}
&I_1 \text{ and } I_2 \text{ construct a partition of the index set } \{1, 2, \dots, n\} \\
&a \in \{0, 1\}, \\
&s_i^a = A_i^a (1 - A_i)^{1-a}, \\
&r_i^a = 1 - \frac{s_i^a}{(\hat{\pi}(X_i))^a (1 - \hat{\pi}(X_i))^{1-a}}, \\
&R_i^a = Y_i - \hat{\mu}(a, X_i), \\
&Z_i = \text{transform}(X_i), \\
&\hat{\Omega}^a = \left(\frac{1}{|I_2|} \sum_{i \in I_2} s_i^a Z_i Z_i^\top \right)^{-1},
\end{aligned}$$

When sample splitting in K folds $(I_1, I_2, \dots, I_K)$, then For j in $(1 : K)$: denote sample in I_j are estimation sample, no in I_j are train sample. using the train sample to calculate $\hat{\Omega}$, and then using the trained $\hat{\Omega}$ and the estimation sample to calculate HOIF_m^a .

3 The Integrated Algorithm (Main Interface)

The whole function combines all the following steps to get the HOIF estimators for ATE. It serves as the primary entry point, orchestrating the data transformation, residual calculation, and the choice of cross-fitting strategy.

Function Inputs:

- Full observed data $(X_i, A_i, Y_i)_{i=1}^n$.
- Nuisance function estimators $(\hat{\mu}(1, X_i), \hat{\mu}(0, X_i), \hat{\pi}(X_i))$.
- Transformation method and its tuning parameters.
- Inverse method of weighted Gram matrix.
- Maximum HOIF order m and `ustat` backend.
- Switch: A boolean flag for sample splitting and the number of splits K .

Procedure:

- (i) Global Pre-processing:
 - Transform the covariates X_i on the whole data to obtain basis functions $(Z_i)_{i=1}^n$.
 - Compute the global residuals $((R_i^1, r_i^1), (R_i^0, r_i^0))_{i=1}^n$.

(ii) Branching Logic:

- **If not using sample splitting:** Proceed with the entire dataset using the steps described in the following sections. Output $(\text{BC}_l^{\text{ATE}}, \text{HOIF}_l^a, \text{IIF}_l^a)$ for $l = 2, \dots, m$ and $a \in \{0, 1\}$ (returned by the package as `ATE`, `HOIF1/HOIF0`, `IIF1/IIF0`).
- **If using sample splitting (Cross-fitting):** a. Split the indices $\{1, \dots, n\}$ into K disjoint parts $(I_1, I_2, \dots, I_K)$. b. For each fold $j = 1, \dots, K$:
 - Training: Use data with indices not in I_j ($i \notin I_j$) to compute the inverse Gram matrices $(\Omega_{1,j}, \Omega_{0,j})$.
 - Estimation: Use data with indices in I_j ($i \in I_j$) and the pre-computed $(\Omega_{1,j}, \Omega_{0,j})$ to compute:
 - Local projection matrices $(B_{1,j}, B_{0,j})$.
 - Local HOIF bias-correction terms $(\text{BC}_{l,j}^{\text{ATE}}, \text{HOIF}_{l,j}^a, \text{IIF}_{l,j}^a)$ for $l = 2, \dots, m$.

c. Aggregation: Average the results across all K folds:

$$\text{BC}_l^{\text{ATE}} = \frac{1}{K} \sum_{j=1}^K \text{BC}_{l,j}^{\text{ATE}}, \quad \text{HOIF}_l^a = \frac{1}{K} \sum_{j=1}^K \text{HOIF}_{l,j}^a, \quad \text{IIF}_l^a = \frac{1}{K} \sum_{j=1}^K \text{IIF}_{l,j}^a$$

Output: Final averaged bias-correction terms $(\text{BC}_l^{\text{ATE}}, \text{HOIF}_l^a, \text{IIF}_l^a)$ for $l = 2, \dots, m$.

4 Step-by-Step Technical Notes

4.1 Transformation of Covariates

First, we transform the covariates X_i into a set of basis functions $Z_i \in \mathbb{R}^k$. Common choices include B-splines or Fourier basis functions.

Function Requirement:

- **Input:** Covariate matrix X , the transformation method (e.g., Fourier or B-splines), and relevant tuning parameters (including the basis dimension k).
- **Output:** The transformed basis matrix $Z \in \mathbb{R}^{n \times k}$, where each row Z_i corresponds to the i -th observation.

4.2 Compute the Residuals

Next, we compute two pairs of residuals, indexed by the treatment assignment $a \in \{0, 1\}$.

For the treatment group ($a = 1$):

$$R_i^1 = Y_i - \hat{\mu}(1, X_i)$$

$$r_i^1 = 1 - \frac{A_i}{\hat{\pi}}(X_i)$$

For the control group ($a = 0$):

$$R_i^0 = Y_i - \hat{\mu}(0, X_i)$$

$$r_i^0 = 1 - \frac{1 - A_i}{1 - \hat{\pi}}(X_i)$$

(The treatment indicator s_i^a is **not** folded into R_i^a here; it enters through the projection matrix B^a below, matching the implementation.)

Function Requirement:

- **Input:** Observed data (A, Y) and estimated nuisance functions $(\hat{\mu}(1, X), \hat{\mu}(0, X), \hat{\pi}(X))$.
- **Output:** Two residual pairs $((R_i^1, r_i^1))_{i=1}^n$ and $((R_i^0, r_i^0))_{i=1}^n$.

4.3 Compute the Inverse of the Weighted Gram Matrix

Compute the inverse of the weighted Gram matrix G_a for each $a \in \{0, 1\}$:

$$G_a = \frac{1}{n} \sum_{i=1}^n s_i^a Z_i Z_i^T,$$

where $s_i^a = A_i^a(1 - A_i)^{1-a}$ serves as the indicator for the a -th group. Let $\Omega_a = G_a^{-1}$ denote the corresponding inverse matrix.

Several estimation methods can be employed for Ω_a , such as direct inversion (e.g., using `chol2inv()` in R for efficiency) or shrinkage estimators (e.g., via the `nlshrink` or `corpcor` R packages) to improve numerical stability in high-dimensional settings.

Function Requirement:

- **Input:** Basis functions Z , treatment indicators A , and the inversion method (`direct`, `nlshrink`, or `corpcor`).
- **Output:** A pair of inverse matrices (Ω_1, Ω_0) .

4.4 Compute the Projection Matrix

Compute the projection-like basis matrix B^a for each $a \in \{0, 1\}$:

$$B^a = Z \Omega_a (s^a \odot Z)^T, \quad \text{i.e.} \quad B_{ij}^a = Z_i^\top \Omega_a s_j^a Z_j,$$

where $Z \in \mathbb{R}^{n \times k}$ is the matrix of basis functions and $s^a \odot Z$ multiplies row j of Z by s_j^a (in code: `B1 = Z %*% Omega1 %*% t(Z * A)`). Note that B^a is an $n \times n$ matrix carrying the treatment indicator on its second index.

Function Requirement:

- **Input:** Basis matrix Z and the inverse matrices (Ω_1, Ω_0) .
- **Output:** Basis matrices (B^1, B^0) .

4.5 Compute the HOIF Estimators

Finally, we compute the HOIF estimators for the ATE.

Function Requirement:

- **Input:**
 - Maximum order m .
 - Residual pairs $((R_i^a, r_i^a))_{a \in \{0,1\}}$.
 - Projection matrices (B^1, B^0) .
 - Backend for `ustat` (`numpy` or `torch`).
- **Procedure:** For each order j from 2 to m , and for each treatment assignment $a \in \{0, 1\}$, calculate the U -statistics:

$$U_j^a = (-1)^j \text{ustat} \left(\text{tensors} = T_j^a, \text{expression} = E_j^a, \text{backend} = \text{backend}, \text{average} = 1 \right)$$

The tensors T_j^a and index expressions E_j^a are defined as:

$$T_j^a = \left(r^a, \underbrace{B^a}_{j-1 \text{ times}}, R^a \right)$$

$$E_j^a = (1, (1, 2), \dots, (j-1, j), j)$$

so that the first tensor r^a is indexed by i_1 and the last tensor R^a by i_j , matching the target formula.

Then, for each order $l \in \{2, \dots, m\}$, compute:

$$\begin{aligned} \text{IIF}_l^a &= \sum_{j=2}^l \binom{l-2}{l-j} U_j^a \\ \text{HOIF}_l^a &= \sum_{j=2}^l \text{IIF}_j^a \\ \text{BC}_l^{\text{ATE}} &= \text{HOIF}_l^1 - \text{HOIF}_l^0 \end{aligned}$$

- **Output:** $(\text{BC}_l^{\text{ATE}}, \text{HOIF}_l^a, \text{IIF}_l^a)$ for $l = 2, \dots, m$, returned by the package as `ATE`, `HOIF1/HOIF0` and `IIF1/IIF0`.

Bibliography

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